

RAG vs Fine-Tuning: Decision Matrix for Enterprise LLMs

CHEAT SHEET | 1 OF 2

Ten criteria compared across RAG, LoRA fine-tuning and the hybrid approach. Ratings are qualitative and every number is illustrative.

CRITERION	RAG (retrieval-augmented generation) Retrieve documents, then answer	Fine-tuning with LoRA LoRA (low-rank adaptation, a parameter-efficient fine-tuning method). Changes behavior, not facts	Hybrid MATURE DEFAULT RAG for facts + light LoRA for behavior
1 Knowledge freshness	Update a document, re-embed it, and it is typically live within minutes.	Knowledge is frozen at training time; retrain to change it.	RAG supplies the facts, so they stay current.
2 Citations and auditability	Every answer can cite its source chunk and page.	No verifiable source to point to; hard to trace why it answered.	Cited facts, delivered in a consistent format.
3 Per-user access control	Filter chunks by permission at retrieval time, before the LLM sees them.	Shared weights cannot reliably enforce per-user access inside the model.	Enforce permissions in retrieval, exactly as with RAG.
4 Hallucination on facts	Reduced when retrieval is good; answers are grounded in retrieved text.	Can memorize wrong or outdated facts and state them confidently.	Reduced when retrieval is good; still validate outputs, since a tuned model can state ungrounded answers.
5 Tone, style, output format	Prompt-only control; output can vary from run to run.	Typically strong, consistent control of tone, JSON and tool-calling format.	Best of both: grounded facts, consistent behavior.
6 Latency	Extra retrieval step; illustrative 0.8–2 s per query.	No retrieval step; illustrative 0.3–0.6 s per query.	Similar to RAG, since it still retrieves first.
7 Cost profile	Ongoing: vector DB, embeddings, longer prompts. Scales with data and traffic.	Upfront training (illustrative: LoRA \$500–\$5K, full fine-tune \$40K–\$100K+ per run); typically low marginal cost.	Both: RAG running costs plus an upfront LoRA training cost.
8 Maintenance	Keep documents, chunking and index fresh; monitor retrieval quality.	Retrain when behavior drifts; run regression evals each time.	Both loops: the retrieval pipeline and the fine-tune.
9 Sensitive-data risk	Source documents stay in your store; only retrieved chunks enter prompts (check where your LLM runs). Delete a document and re-index and it is no longer retrieved; also purge logs, caches and backups.	Training data is diffused into the weights and is hard to remove.	Keep sensitive facts out of the training data; retrieve them instead.
10 Best for	Changing knowledge, cited answers, compliance.	Consistent behavior, output format or a specific skill.	Most enterprise assistants: facts from RAG, behavior from a light LoRA.

RATING Strong fit Mixed: depends on your setup Weak fit or main risk

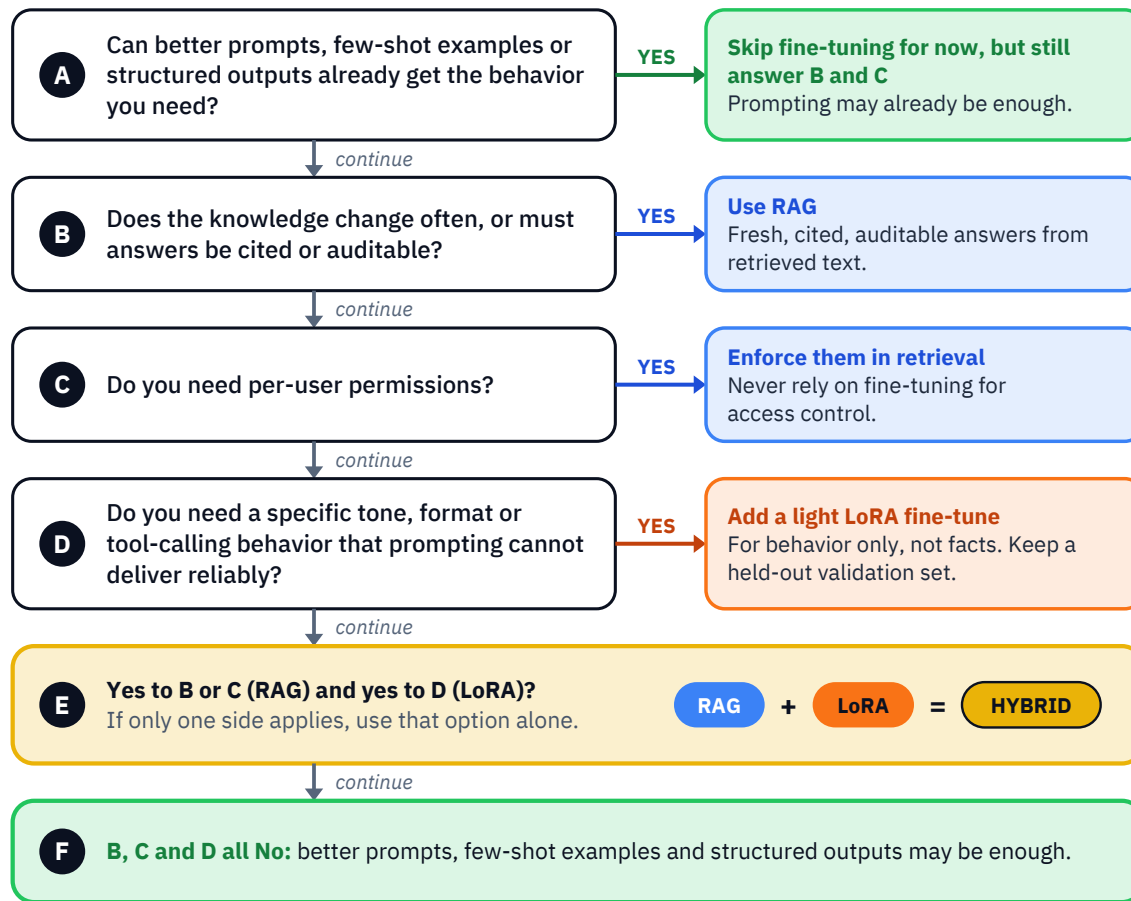
All numbers are illustrative ranges, not benchmarks. Measure latency, cost and quality on your own stack. Ratings are qualitative judgments of fit for an enterprise assistant.

How to decide, and what to avoid

CHEAT SHEET | 2 OF 2

A short decision flow, the failure modes of each approach, and a checklist to run before you build.

Decision flow: answer in order



Where RAG fails

- ✗ **Chunking mid-sentence.** Poor chunking can noticeably degrade retrieval (the video cites 30–40% as an illustrative figure): measure yours.
- ✗ **No re-ranking.** For example, retrieve the top 50, re-score, keep the best 5.
- ✗ **Stale embeddings.** Documents change but the index does not.
- ✗ **No permission filtering.** An intern's query could surface executive-compensation chunks.
- ✗ **No retrieval eval set.** Without one you cannot tell if retrieval is improving.

Where fine-tuning fails

- ✗ **Overfitting a narrow dataset.** The model can memorize examples and generalize poorly.
- ✗ **Catastrophic forgetting.** Training on a narrow task can degrade general abilities.
- ✗ **Teaching facts instead of behavior** (fine-tuning is a poor way to add or update facts). They go stale and have no verifiable source.
- ✗ **No held-out validation set.** You cannot tell whether the model generalizes or overfits.
- ✗ **Sensitive data leaking into weights.** Hard to remove once it has been trained in.

Before you build

Six steps, in order

- 1 Write 20–50 real questions, each with its expected source.
- 2 Start with RAG plus good prompts.
- 3 Measure retrieval hit rate before touching the model.
- 4 Add re-ranking and citations.
- 5 Only then consider LoRA for tone or format, with a held-out validation set.
- 6 Re-run regression evals after every change.